

From local to global almost-everywhere constancy

MathlibAnnex.exists_aeConstantOn_of_local

theorem

Separates agreement of constants by connectedness from countable measure-theoretic gluing.

Statement

Let X be a topological space with a measurable structure, and let μ be positive on every nonempty open set. Let Y be any set, $u : X \rightarrow Y$, and let $U \subseteq X$ be connected and Lindelöf. Suppose that for every $x \in U$ there are an open neighborhood V_x and a constant $c_x \in Y$ such that

$$x \in V_x \subseteq U, \quad u = c_x \quad \mu|_{V_x}\text{-almost everywhere.}$$

Then there is one $c \in Y$ with $u = c$ almost everywhere for $\mu|_U$.

Assumptions

Here connected means nonempty and admitting no partition into two disjoint nonempty relatively open subsets. Lindelöf means that every open cover of U admits a countable subcover. No finiteness or Haar assumption is imposed on μ , and X need not be second countable. No topology or measurable structure on the value set Y is required. The supplied neighborhoods in particular make U open, although openness is not a separate input.

Conclusion

One common constant works outside a null set for the restricted measure on U . Connectedness identifies the local constants; Lindelöf supplies the countability needed to join their exceptional sets. Those are distinct uses of the hypotheses.

The local data at x are an open set V , the relations $x \in V \subseteq U$, and a value $c \in Y$ satisfying $u = c$ almost everywhere for $\mu|_V$. At one chosen point of the nonempty set U , these data supply the initial value $c_0 \in Y$. Thus no separate nonemptiness assumption on Y is needed.

Proof route

Fix a constant from one local neighborhood. Its region of local validity and its complement in U are disjoint open sets. Connectedness makes that region all of U . Select a countable subcover of neighborhoods carrying this same constant and apply countable gluing.

▼ Proof steps

1. **Fix a value and its region.** Choose $x_0 \in U$ and its local data (V_0, c_0) . Define

$G := \{x \in U : \exists V \text{ open, } x \in V \subseteq U, u = c_0 \mu|_V\text{-almost everywhere}\}.$

It contains x_0 . It is open: a witnessing V for x is contained in G , because the same V witnesses membership for every point of V .

2. **Show the complementary region is open.** For $x \in U \setminus G$, choose local data (V_x, c_x) . If some $y \in V_x$ were also in G , take a witnessing open set W for y with constant c_0 . Then $y \in V_x \cap W$, so this open overlap has positive measure. The open-overlap theorem gives $c_x = c_0$. But (V_x, c_x) would then witness $x \in G$, a contradiction. Thus

$$x \in V_x \subseteq U \setminus G,$$

and $U \setminus G$ is open.

3. **Use connectedness to identify the region.** The disjoint open sets G and $U \setminus G$ cover U . Preconnectedness forces U to lie in one of them. It cannot lie in the latter because $x_0 \in G \subseteq U$. Hence $G = U$. In particular, every point of U now has an open neighborhood on which the same c_0 works almost everywhere.
4. **Use a countable cover to glue.** Choose for each $x \in U$ one of these neighborhoods V_x . Lindelöf gives a countable set $T \subseteq U$ of indices such that

$$U \subseteq \bigcup_{x \in T} V_x, \quad u = c_0 \mu|_{V_x}\text{-almost everywhere for each } x \in T.$$

The countable-cover gluing theorem gives $u = c_0$ almost everywhere for $\mu|_U$. Only this selected countable family is used in the measure argument; the original uncountable family of local null sets is never united.

Main citations

- Exact declaration and its source — `MathlibAnnex.exists_aeConstantOn_of_local`
- Agreement of constants on a positive open overlap — `MathlibAnnex.aeConstants_eq_of_open_overlap`
- Countable gluing for restricted measures — `MathlibAnnex.aeConstantOn_of_countableCover`
- Almost-everywhere equality for a restricted measure — `MathlibAnnex.AEConstantOn`
- The six-field local almost-everywhere constancy datum — `MathlibAnnex.LocalAEConstantAt`
- The region carrying one specified local constant — `MathlibAnnex.constantRegion`
- Openness of that constant region — `MathlibAnnex.isOpen_constantRegion`
- Lindelöf selection followed by countable gluing — `MathlibAnnex.aeConstantOn_of_everywhere_local`
- Openness of the complementary constant region — `MathlibAnnex.isOpen_constantRegion_compl`
- Preconnectedness propagates a nonempty constant region — `MathlibAnnex.constantRegion_eq_of_preconnected`

Lean source signature (exact)

```
theorem exists_aeConstantOn_of_local
  {U : Set X} (hU : IsConnected U) (hL : IsLindelof U) {u : X → Y}
  (hlocal : ∀ x ∈ U, Nonempty (LocalAEConstantAt μ U u x)) :
  ∃ c : Y, AEConstantOn μ u U c
```

In the source	Mathematical meaning
<code>[MeasurableSpace X] [TopologicalSpace X] (μ : Measure X) [μ.IsOpenPosMeasure]</code>	The topological measure space with positivity on nonempty open sets, and arbitrary value set Y .
<code>{U : Set X} (hU : IsConnected U)</code>	The domain U is nonempty and preconnected. This exact theorem excludes the empty domain.
<code>(hL : IsLindelof U) {u : X → Y}</code>	Every open cover of U has a countable subcover, and u takes values in Y . Second countability of all X is not required.
<code>(hlocal : ∀ x ∈ U, Nonempty (LocalAEConstantAt μ U u x))</code>	At every $x \in U$ there exists a local record with the six field rows below. Its constant may initially depend on x .
<code>neighborhood : Set X</code>	The selected neighbourhood V of the point x ; this and the next five rows read the separately cited local record.
<code>isOpen_neighborhood</code>	V is open in the ambient topology.
<code>mem_neighborhood</code>	The specified point satisfies $x \in V$.
<code>neighborhood_subset</code>	$V \subseteq U$.
<code>constant</code>	One value $c \in Y$ for this neighbourhood.
In the source	Mathematical meaning
<code>ae_eq</code>	$u(y) = c$ for $(\mu _V)$ -almost every y . The same V, c occur in these six fields.
<code>∃ c : Y, AEConstantOn μ U U c</code>	The conclusion supplies one value c with $u = c$ almost everywhere for $\mu _U$. Connectedness identifies constants, whereas Lindelöf supplies countability for the exceptional sets.

▼ Exact surrounding binder context (separate excerpts)

Exact source lines 12–15:

```
noncomputable section
open Set MeasureTheory Filter TopologicalSpace
open scoped ENNReal Topology
namespace MathlibAnnex
```

Exact source lines 49–52:

```
section Topological
variable {X Y : Type*} [MeasurableSpace X] [TopologicalSpace X]
variable (μ : Measure X)
```

Exact source lines 90–90:

```
variable [μ.IsOpenPosMeasure]
```

▼ **Exact content identity**

Declaration: MathlibAnnex.exists_aeConstantOn_of_local

Accepted content SHA-256: 32eb30dc10b1168e6498f64852c670422dee306e24e529c37903ca272446043d

Accepted source guide SHA-256: 8f2b48a24d4619c2f6a92822335346698edaba2bc7d32939b663da8392d43bae

Source commit: 437e6e46228bbb8e91211ebded349d7a30020e73